are uploaded to the onboard mission computer; during flight, the vehicle autonomously follows and adapts the plan using sensor feedback, health monitoring, and rule-based or algorithmic decision logic.

- *Redundant, fault-tolerant flight control*: Robust operation in contested or harsh environments is achieved through redundant flight control architectures (multiple flight computers, duplicated sensors, cross-monitoring). Flight control software implements fault detection, isolation, and recovery (FDIR), graceful degradation, and automatic fail-safe behaviors (e.g., loiter, return-to-home, controlled descent).
- *Networked, multi-layer communications*: Third-generation UAVs employ layered communications (line-of-sight RF links, SATCOM, relay via other platforms) so that command, control, and high-bandwidth sensor data can be exchanged across theatre and strategic networks. Networked operations enable distributed command-and-control, sensor fusion across platforms, and real-time tasking from remote users.
- *Multi-sensor data fusion and onboard processing*: Multiple sensors (EO/IR, SAR, SIGINT, AIS, LiDAR, etc.) provide complementary situational awareness. Onboard data processing and fusion reduce latency by performing feature extraction, automatic target recognition (ATR), geo-registration, and compression before transmitting condensed intelligence to operators or higher-level networks.
- *Persistent, beyond-line-of-sight (BLOS) operations*: SATCOM and autonomous flight enable operations far beyond visual range for extended periods (MALE/HALE endurance). Persistence is a defining principle: continuous presence over areas of interest to deliver time-critical intelligence or persistent deterrence.